EVALUATION REPORT / BUILD OF 24 OCT 2026

BANTAY-BAHA

Flood-linked dengue surge risk and storm-driven loss of hospital access, ranked by province for any typhoon track. Results of the full pipeline run on 501 verified source files.

PJDSC 2026

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WHAT WE BUILT, WHAT THE DATA SAYS

Bantay-Baha joins three graphs into one ranking: a flood-exposure layer (rain, river routing, flood hazard), a lead-lag outbreak graph learned with PCMCI+, and a road-plus-ferry access graph that a typhoon cuts. A quantile forecast of dengue gives the probability **R** that a province's next four weeks exceed its usual bad month; the access model gives the share **A** of people cut off from a hospital; priority is **P = R × A**. Every number below was computed by `build.bat` from the raw files; nothing is synthetic.

0.55
relative WIS vs seasonal-naive, 1-4 weeks (target below 1)

85%
90% interval coverage (target 80-95%)

8
PCMCI+ lead-lag links among 84 provinces (cross-correlation: 4,265)

60
pre-computed scenarios (4 typhoons x 15 tracks)

FINDINGS

- **The forecast beats the bar.** Pooled multi-quantile XGBoost reaches relative WIS 0.42 at one week and 0.66 at four weeks over 34,170 rolling-origin forecasts in 2018-2019; 90% intervals cover 85%.
- **Two honest null results.** Lead-lag graph features do not improve the forecast (relative WIS graph / no-graph = 0.999), and adding rain, river and flood-hazard features moved relative WIS only from 0.547 to 0.545. At 1-4 weeks, recent cases and the season carry the skill.
- **Surge probability is informative but under-confident.** Brier 0.089 vs 0.216 for the base rate; forecasts of 0.1-0.7 come true 10-20 points more often. It needs recalibration before the tier thresholds are trusted.
- **The ranking points at the right places.** Spearman(R, observed next-4-week incidence) = 0.69 for Ompong and 0.74 for Tisoy; 7 of Ompong's top-10 provinces by priority surged against a 40% base rate.
- **Access loss is large and robust to settings.** The typhoons' gale swaths cancel 42-100 of 108 RoRo routes and cut off 0.3 M-1.0 M people (roads cut by Sentinel-1 observed flood); Kendall's τ of the ranking under nine setting changes is at least 0.99.

> **Not yet in these results:** the MAP friction raster access method (data on disk, not wired into the scenarios); the DOH NHFR hospital list and PSA PSGC (manual downloads); the Chronos-2 / TabPFN-TS benchmarks. See the scorecard (page 13).

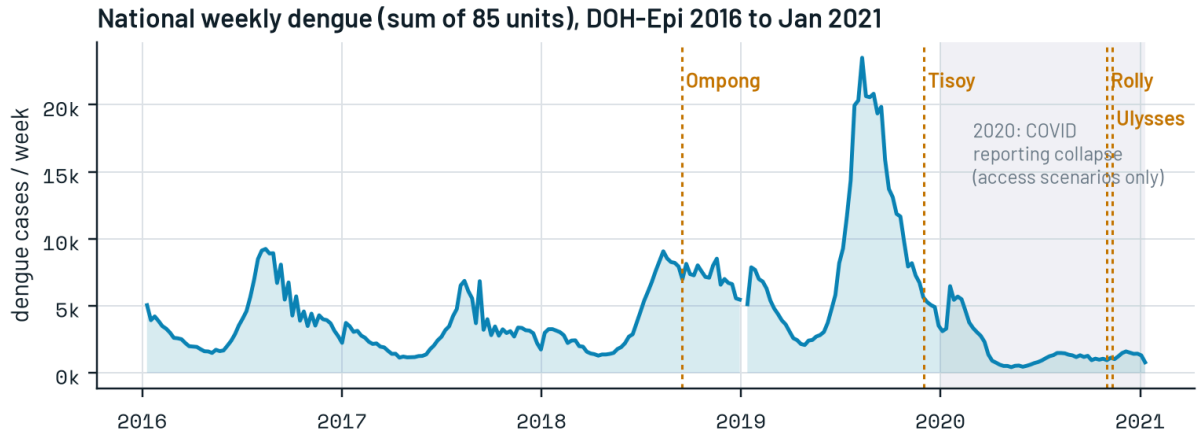


Figure 1. The forecast target: DOH-Epi weekly dengue summed over the 85 harmonised units, with the four backtest typhoons. 2020 is used for access scenarios only.

SUMMARY

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DATA AND ACQUISITION

501 files from 22 sources (7.0 GB) are listed in `data/raw/MANIFEST.csv` with size and sha256; all were verified on disk before this build. Open sources come from `scripts/download_data.py`, Earth Engine sources from `scripts/download_gee.py`, and the team shares them through a private bucket (`setup.bat`, every file checked against the manifest).

FOLDER	DATASET	ROLE	FROM	FILES	MB
dengue_hdx	DOH-Epi dengue, province/city x week 2016–Jan 2021	forecast target	HDX	1	2
census2020	2020 CPH population by municipality	population	HDX	1	1
cod_ab	COD-AB boundaries ADM0-4 (v03, 2025)	join keys, geometry	HDX	2	370
ibtracs	IBTrACS v04r01 Western Pacific	storm exposure, swaths	NOAA NCEI	1	115
chirps	CHIRPS v2.0 daily 0.05°, 2016–2020	rain features, routing	IRI Data Library	5	563
imerg	GPM IMERG V07 daily 0.1°	rain features	Earth Engine	60	68
hydrorivers	HydroRIVERS v1.0 Australasia	river routing	HydroSHEDS	2	139
noah	UP NOAH flood hazard 5/25-yr, 139 provinces	static susceptibility, road proxy	UP NOAH (Drive)	142	4,733
copernicus_ems	EMSR312 / EMSR478 vector packages	observed flood (Ompong AOIs)	Copernicus EMS	16	28
worldpop	WorldPop 2020 100 m UN-adjusted	population grid	WorldPop	1	159
meta_hrsl	Meta HRSL under-5 / 60+	vulnerability (not yet used)	HDX	2	112
osm	OpenStreetMap Philippines PBF	road network	Geofabrik	1	611
roroph	roroph RoRo routes (108)	ferry edges (2024–26 proxy)	pinasr/roroph	2	<1
hotosm_health	HOT OSM health facilities	hospital nodes (fallback)	HDX	1	2
noah_facilities	UP NOAH critical facilities	hospital nodes (fallback)	UP NOAH	1	4
map_friction	MAP friction surface 2019	raster access (wired next)	Earth Engine	2	2
gfd	Global Flood Database events	context (ends 2018)	Earth Engine	1	<1
sentinel1	Sentinel-1 scene inventory	flood-mapping plan	Earth Engine	1	<1
cchain	Project CCHAIN PIDSR city-week	leptospirosis (stretch)	HDX	4	7
mendeley_parreno	Parreño national weekly notifiable diseases	cross-check	Mendeley Data	1	<1
opendengue	OpenDengue V1.3	national cross-check	OpenDengue	3	110
s1_flood	*Sentinel-1 flood masks per typhoon (UN-SPIDER)*	*observed flood*	*Earth Engine*	–	*pending*
psgc	*PSA PSGC 2Q 2026*	*pcode check*	*manual*	–	*pending*
nhfr	*DOH NHFR*	*hospital list*	*manual*	–	*pending*

STEP 0: ONE UNIT-WEEK TABLE

HARMONISED UNITS

The DOH file has **126 reporting units, not 81 provinces**: provinces mixed with 45 highly urbanised and component cities, plus misspellings (NUEVA VISCAYA, SISQUIJOR, BASILIG CITY = Bislig). COD-AB uses current boundaries (Maguindanao split in 2022, Special Geographic Area in 2024). The reviewed crosswalk in `data/crosswalks/` folds cities into their province and undoes the post-2020 changes, giving **85 units**; all 126 dengue units resolve and every unit has data. A unit-week counts only if every member reported (NaN otherwise, never 0).

POPULATION

The 2020 census table's municipal codes are old 9-digit PSGC and do not join to COD-AB for 54 municipalities (24.2 M people). Its own *Province* column already places cities in their province, so units are summed on it: 108.67 M people.

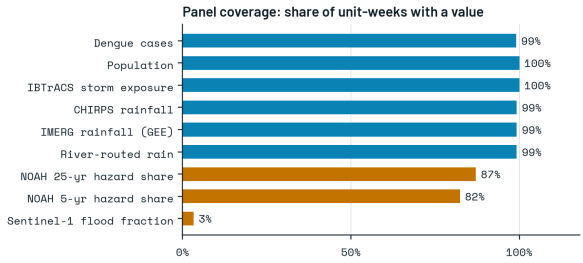


Figure 2. Coverage of the 85 x 262 panel. Sentinel-1 is filled only for the four landfall windows (4 weeks each), by design.

LAYER 1 INPUTS

Rain: CHIRPS and IMERG agree (weekly r = 0.87). River routing accumulates CHIRPS rain down the HydroRIVERS network. NOAH 25-yr hazard covers 74 of 85 units; the median unit has 20% of its people in a hazard zone. Units with no map are **missing, not zero**: Basilan, Batanes, Biliran, Camiguin, Davao Occidental, Guimaras, Northern Samar, Romblon, Siquijor, Sulu, Tawi-Tawi.

Storm exposure: every Western Pacific track 2016-2020 from IBTrACS, gale radius from USA R34, else JMA R30, else a fixed 150 km (the fallback never triggered for the four backtest storms).

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METHODS AND EVALUATION PROTOCOL

LAYER 2: LEAD-LAG OUTBREAK GRAPH

log(1 + incidence), 52-week seasonal difference, standardised inside the training window. PCMCI+ (Runge, 2020) with partial-correlation tests, lags 1-3 weeks, pc_alpha 0.01; lagged inter-province edges are kept when the Benjamini-Hochberg (1995) q-value is below 0.05. The FDR step matters: on pure noise with 85 series, PCMCI+ alone returned 60-90 false edges. PageRank (Page et al., 1999) on the reversed graph scores sentinels; Louvain (Blondel et al., 2008) finds communities; lagged cross-correlation with the same FDR is the baseline graph.

FORECAST

One pooled XGBoost (Chen & Guestrin, 2016) per horizon (1-4 weeks) with the multi-quantile objective at the 23 CDC FluSight levels, on log(1 + incidence), back-transformed to cases. Features: lagged incidence and moving averages, same week last year, week of year, CHIRPS / IMERG / river rain lags, storm exposure, NOAH hazard, and leader signals from the graph. SHAP values (Lundberg & Lee, 2017) from the median model give the dashboard's "drivers". A second model predicts the 4-week cumulative incidence directly; **R** is its probability of exceeding the unit's 90th percentile of 4-week incidence in the training window.

EVALUATION

Rolling origin, never a random split. From January 2018 to December 2019 the model, the lead-lag graph and the surge thresholds are refit every 13 weeks on data up to the refit date only, then forecast every origin until the next refit. Training targets never pass the refit date and 2020 is never a target. Scores: weighted interval score on case counts (Bracher et al., 2021), relative WIS against a seasonal-naive forecast with empirical error quantiles, 50% and 90% coverage, and an online conformal 90% band built from past errors only (Gibbs & Candès, 2021) as a check on the quantiles.

LAYER 3: ACCESS

OSM highways (motorway to track) become a graph of junctions; RoRo routes are ferry edges (crossing time plus half the headway); hospitals are HOT OSM plus NOAH name-matched facilities (2,199) until NHFR arrives. WorldPop (Tatem, 2017) is summed to 1 km cells, each joined to the nearest road node. A multi-source Dijkstra gives hours to the nearest hospital. A storm cancels ferry routes crossing its gale swath and cuts roads in observed flood (Sentinel-1, observed tracks) or, for shifted tracks, cuts (high hazard) or slows (medium) roads in NOAH 25-yr hazard inside the swath. A cell is cut off if it loses its route, or its trip at least doubles and grows by more than an hour (after Petricola et al., 2022).

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FORECAST SKILL

HORIZON	N	WIS XGB	WIS NO-GRAPH	WIS NAIVE	REL WIS	GRAPH / NO-GRAPH	COV 50	COV 90	COV 90 CONFORMAL
1 wk	8,670	14.8	14.8	35.1	**0.42**	1.000	0.46	0.89	0.88
2 wk	8,585	18.0	18.1	35.4	**0.51**	0.999	0.44	0.86	0.87
3 wk	8,500	21.1	21.1	35.6	**0.59**	0.999	0.42	0.84	0.86
4 wk	8,415	23.7	23.7	35.9	**0.66**	0.999	0.40	0.82	0.85
all	**34,170**	**19.4**	**19.4**	**35.5**	**0.55**	**0.999**	**0.43**	**0.85**	**0.86**

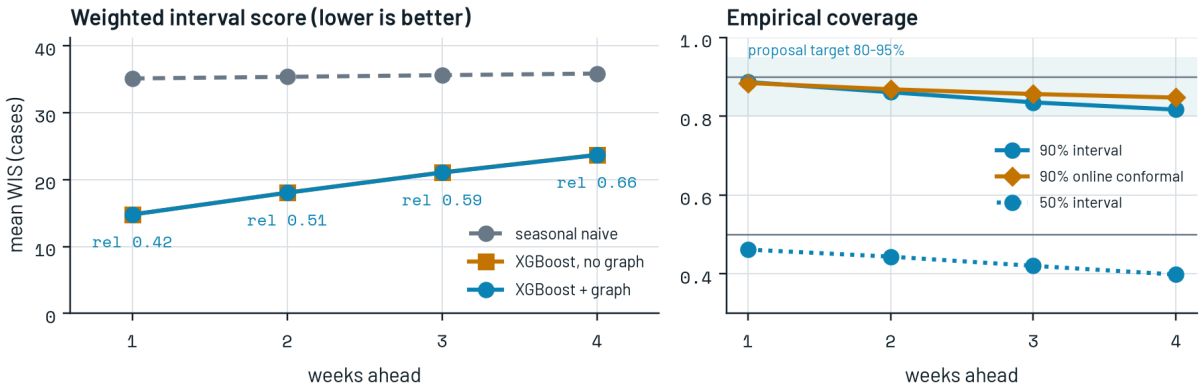


Figure 3. Left: mean WIS by horizon (labels: relative WIS vs seasonal-naive). Right: coverage against the nominal 50% and 90% lines and the proposal's 80-95% band.

Targets met. Relative WIS is below 1 at every horizon (0.42 to 0.66) and 90% coverage is inside 80-95% at every horizon (four weeks is close to the floor at 0.82). The 50% intervals are a little narrow (0.40-0.46). The online conformal band reaches similar coverage, a sign the quantiles are not badly miscalibrated.

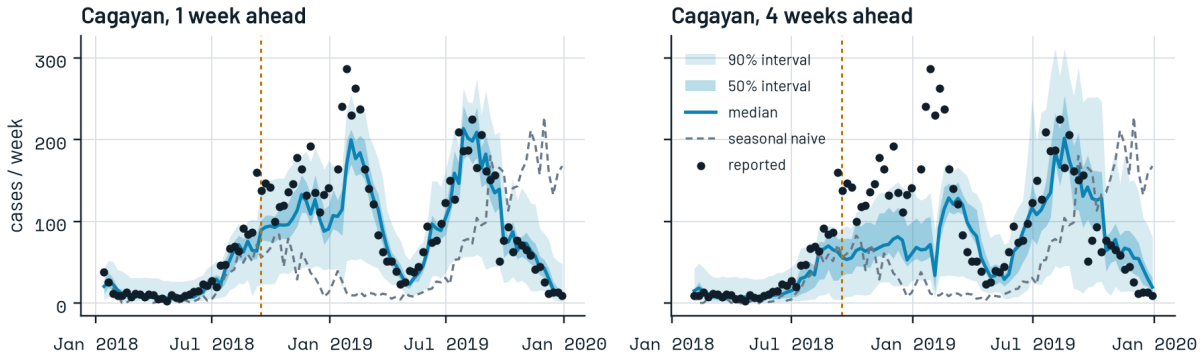


Figure 4. Rolling-origin forecasts for Cagayan (Ompong's landfall province, dashed line), one and four weeks ahead.

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ABLATIONS AND CALIBRATION

WHAT CARRIES THE SKILL

COMPARISON	RELATIVE WIS	READING
XGBoost vs seasonal-naive	0.545	45% lower WIS than "same week last year"
with vs without graph leader features	0.999	no gain: the graph has 0 links in every 2018 window and 2-6 in 2019
with vs without rain / river / hazard (two builds)	0.545 vs 0.547	no meaningful gain at 1-4 weeks

The graph ablation follows from the graph itself: with so few significant links, leader signals are empty for most provinces. The rain result has two likely reasons, not yet tested: rain moves with the season in a four-year panel, so the season features already carry it, and any effect of rain at lags longer than four weeks is outside this forecast horizon. Both are reported as findings rather than tuned away.

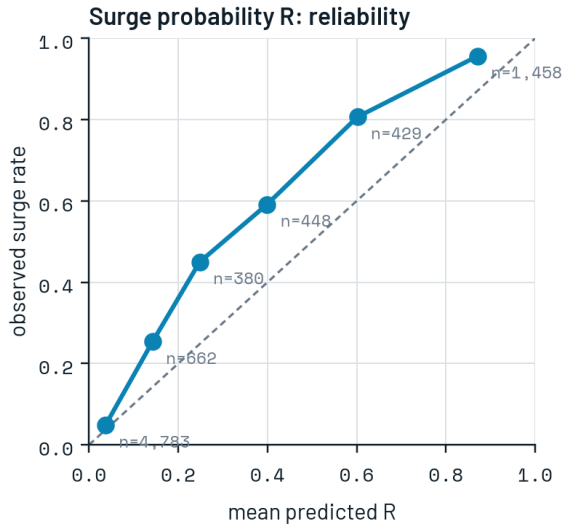


Figure 5. Reliability of R over the 2018-2019 unit-weeks. Points above the diagonal: R is under-confident.

SURGE PROBABILITY R

Brier score 0.089 against 0.216 for always predicting the base rate. The base rate is 32%, not 10%, because 2019 was an epidemic year and the thresholds come from earlier, quieter years.

R is under-confident in the middle range: forecasts of 0.1-0.7 come true 10-20 points more often. Isotonic recalibration fitted on past rolling-origin forecasts only is the next step, before the tier thresholds (R ≥ 0.3, R ≥ 0.5) are relied on.

ABLATION

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THE LEAD–LAG GRAPH



Final window: 84 series, T = 156 weeks after seasonal differencing (Davao Occidental sits out with more than 10% missing weeks). PCMCI+ keeps **8** lagged inter-province links at q < 0.05. Plain lagged cross-correlation with the same FDR keeps **4,265**, because the 2019 national epidemic makes almost every pair co-move.

The proposal's sentinel idea is therefore **weakly supported at province-week level**. The links are statistical dependencies, not evidence of transmission, and several join distant provinces. Likely reasons: weekly province totals average out local waves, four years is short for 84 series, and reporting artefacts dominate small provinces. Synchrony studies such as van Panhuis et al. (2015) worked at longer time scales.

Figure 6. The 8 PCMCI+ links learned on 2016–2019 (teal = leader).

LEADER	FOLLOWER	LAG (WK)	PARTIAL CORR	Q	BOOTSTRAP FREQ
Cavite	Sorsogon	1	0.38	0.001	0.70
Metropolitan Manila Second District	Bulacan	3	0.36	0.006	0.70
Quirino	Surigao del Sur	1	0.36	0.007	0.65
Ilocos Norte	Romblon	1	0.33	0.020	0.75
Metropolitan Manila Third District	Zamboanga del Norte	1	0.33	0.029	0.85
Sulu	Ilocos Sur	1	0.32	0.037	0.55
Camarines Sur	Pangasinan	1	0.31	0.042	0.65
Samar (Western Samar)	Antique	1	0.31	0.045	0.55

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OBSERVED FLOOD: SENTINEL-1

Radar sees through typhoon clouds. For each typhoon, new open water is mapped on Google Earth Engine (Gorelick et al., 2017) by comparing Sentinel-1 VH backscatter before and after landfall, in 100 m tiles near the track: before = median of days -30 to -3, after = darkest image of days 0 to +10, **both from the same relative orbit**, so the viewing geometry matches. A pixel is new flood when VH drops by at least 3 dB and ends water-dark (at most -18 dB); permanent water (JRC), slopes over 5° and specks under 8 pixels are removed (after UN-SPIDER, n.d.).

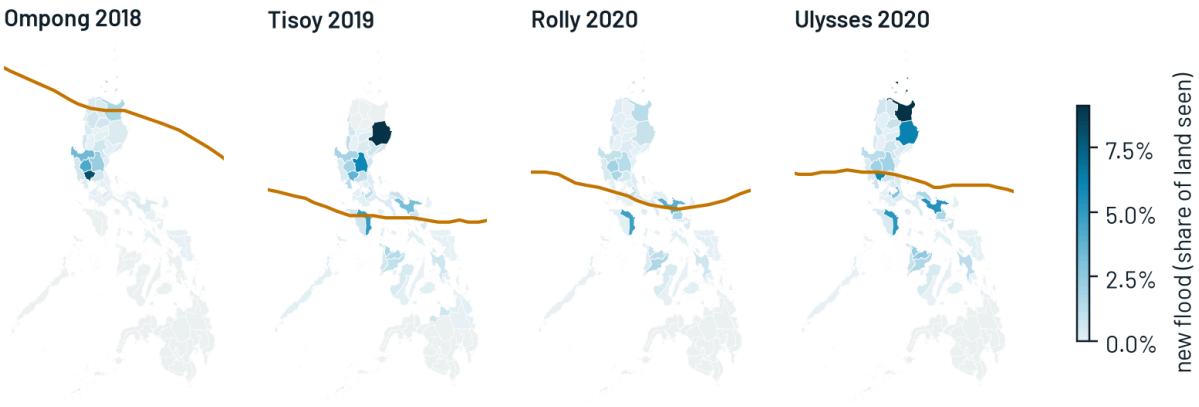


Figure 7. New-flood share of the land Sentinel-1 saw, per unit, in the landfall week, with each track.

TYPHOON	PEOPLE IN NEW FLOOD	UNITS >1% FLOODED	MAX UNIT SHARE	MOST PEOPLE FLOODED
Ompong	0.5 M	5	8.0%	Pampanga
Tisoy	1.1 M	14	9.6%	Isabela
Rolly	0.6 M	10	4.7%	Camarines Sur
Ulysses	1.5 M	15	10.2%	Cagayan

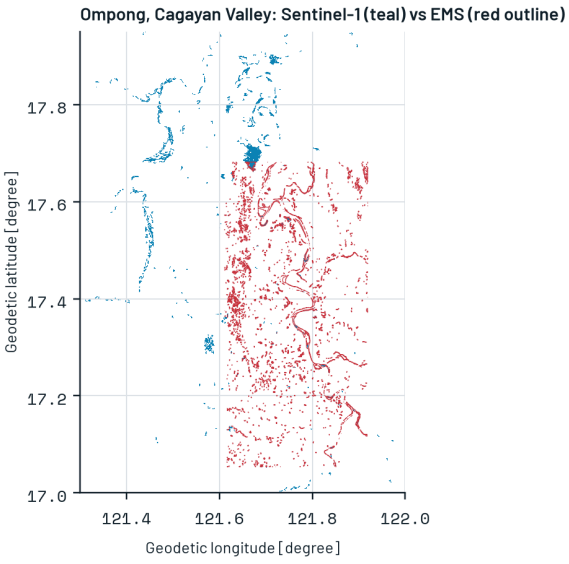


Figure 8. Ompong around Tuguegarao: Sentinel-1 new flood (teal) and Copernicus EMS observed flood (red outlines).

CHECKED AGAINST COPERNICUS EMS

Inside the EMS areas of interest, EMS mapped 108 km^2 of observed flood and Sentinel-1 flagged 46 km^2; 26 km^2 overlap. So **55%** of Sentinel-1 flood there is confirmed by EMS, and Sentinel-1 recovers **24%** of the EMS area.

The first version followed UN-SPIDER's after/before ratio on dB values and recovered almost none of the EMS flood: dividing negative dB values under-detects surfaces that were already dark, and mixing orbits compared different viewing geometries. Sampling backscatter at EMS flood points showed drops of 4-10 dB that the ratio missed. Low recall is expected: EMS used other imagery and dates, while one Sentinel-1 orbit passed here only on 16 and 22 September.

ACCESS LOSS UNDER FOUR TYPHOONS

Baseline. 109.4 M people in 1 km cells; 97.3% have a road or ferry route to one of 2,199 hospitals, with a population-weighted mean trip of 0.24 h. Results are also written for 1,641 municipalities.

TYPHOON	FERRY ROUTES CUT	PEOPLE CUT OFF	MEAN A	MEAN R	DEPLOY / PRE-POSITION	TOP PRIORITY
Ompong Mangkhut 2018	42	0.3 M	0.2%	0.23	0 / 21	Pampanga
Tisoy Kammuri 2019	100	0.7 M	0.6%	0.08	0 / 6	Camarines Norte
Rolly Goni 2020	42	0.4 M	0.3%	0.02	0 / 0	Oriental Mindoro
Ulysses Vamco 2020	54	1.0 M	0.8%	0.04	0 / 2	Cagayan

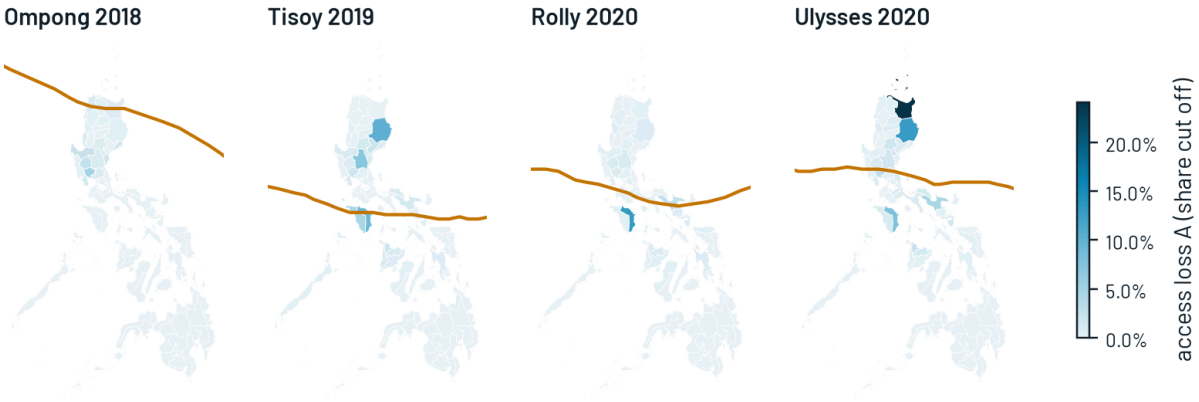


Figure 9. Access loss A per unit for the observed tracks. Roads are cut by Sentinel-1 observed flood; ferries crossing the gale swath are cancelled.

> Roads are cut only where Sentinel-1 saw new water, so cut-offs are storm-specific; the radar misses floods under dense vegetation and in built-up areas and between passes, so these are lower bounds. With the NOAH 25-yr hazard proxy instead (build of 24 Oct, before Sentinel-1), the same tracks cut off 4.8-10.3 M people: an upper bound. The truth lies between. Ferry routes are 2024-2026 operations used as a proxy for 2018-2020. Shifted tracks always use the NOAH hazard proxy (no image of a storm that did not happen).

PRIORITY: OMPONG 2018

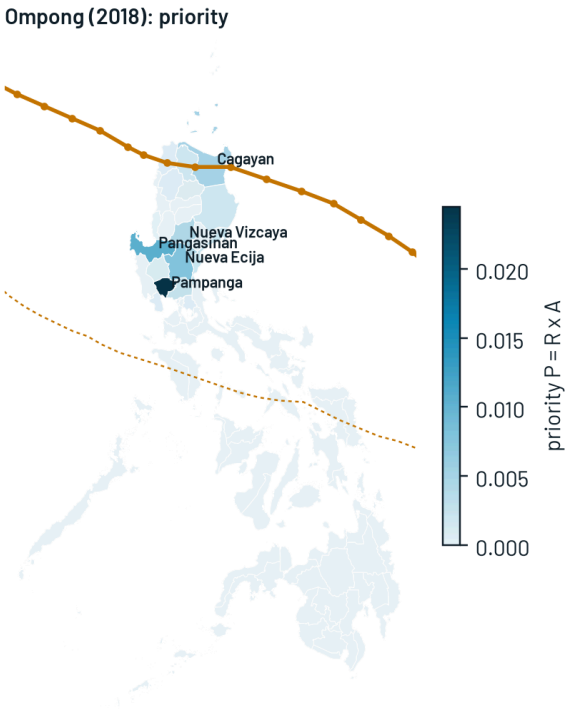


Figure 10. P = R x A for Ompong with its track and gale swath; top 5 labelled.

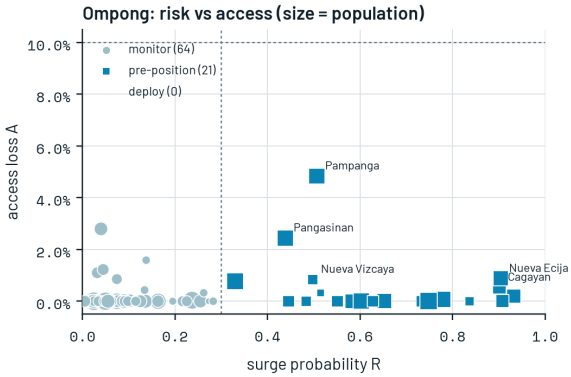


Figure 11. Risk vs access. Dotted lines are the provisional tier thresholds (R = 0.3, A = 10%); marker shape encodes the action tier.

Ompong made landfall in Cagayan on 15 September 2018. The model, trained on data to the landfall week only, puts **Pampanga** first, with 0 provinces in the "deploy mobile clinics" tier and 21 in "pre-position supplies". Provinces with high R but little observed loss of access fall to "pre-position"; "deploy" needs at least 10% of people cut off, which observed flood alone did not produce for this storm (the hazard-map upper bound did, for 7 provinces).

The dashboard repeats this for every storm and 14 shifted variants of each track (±0.5° and ±1° north, ±1° east), so a health officer can ask "what if it comes in further south?" and get an answer in seconds.

BACKTEST, SENSITIVITY, BENCHMARK

DID HIGH-PRIORITY PROVINCES SEE MORE CASES?

TYPHOON	ORIGIN	UNITS	SURGED	P(R, OBS)	P(P, OBS)	TOP-10 SURGED	BRIER R	NOTE
Ompong	2018-09-16	85	40%	0.69	0.21	70%	0.173	
Tisoy	2019-12-08	85	5%	0.74	-0.05	0%	0.040	
Rolly	2020-11-01	61	0%	0.50	0.05	0%	0.001	2020: COVID-depressed reporting, indicative only
Ulysses	2020-11-15	85	1%	0.35	0.12	0%	0.016	2020: COVID-depressed reporting, indicative only

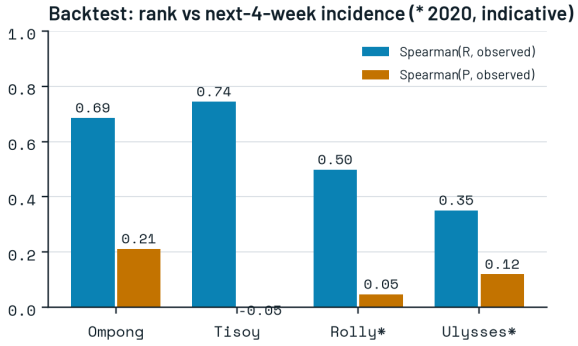


Figure 12. Spearman correlation of each ranking with observed incidence over the next four weeks (relative to each unit's threshold).

R ranks provinces well for both 2018-2019 storms. P correlates less with cases, as it should: A measures access, not disease. After Ompong, 7 of the 10 highest-priority provinces surged against a 40% base rate. After Tisoy almost nothing surged (the 2019 season was over), so the top-10 rate is 0 and the Brier score is low for the right reason.

SENSITIVITY: KENDALL'S τ OF P AGAINST THE BASE SETTINGS

VARIANT	OMPONG τ	TISOY τ	ROLLY τ	ULYSSES τ	MEAN A (OMPONG)
cut-off ratio 1.5	1.000	0.997	1.000	1.000	0.23%
cut-off ratio 3	1.000	0.997	0.999	0.999	0.23%
ferry rule: port in swath	1.000	1.000	1.000	1.000	0.23%
gale fallback 100 km	1.000	1.000	1.000	1.000	0.23%
gale fallback 200 km	1.000	1.000	1.000	1.000	0.23%
medium hazard slow x0.25	1.000	1.000	1.000	1.000	0.23%
medium hazard slow x0.75	1.000	1.000	1.000	1.000	0.23%
speed x0.8	1.000	0.998	0.987	0.998	0.23%
speed x1.2	1.000	0.996	0.986	0.998	0.23%

Every variant changes A (speed ±20% moves mean A by about 4%) but none reorders the ranking: P's spread is dominated by R, which ranges 0.02-0.93 across provinces. The gale fallback never triggers for these storms.

METRO MANILA BENCHMARK

Cutting every NCR road in NOAH 25-yr medium or high hazard leaves **29.7%** of NCR without a route within the cut-off rule. Tariverdi et al. (2023) report about 22% losing access to higher-level care at a 15 cm flood. The definitions differ, so this is an order-of-magnitude check, not a replication.

PROPOSAL OBJECTIVES: SCORECARD

OBJECTIVE (PROPOSAL)	RESULT	STATUS	NOTE
Province-week panel, 2016-2020, joined on PSGC	85 harmonised units x 262 weeks; dengue, rain, river, storms, NOAH, Sentinel-1	met	
Lead-lag graph, sentinels, communities; test sentinel features out of sample	PCMCI+ with FDR: 8 links; ablation run	met	result is null (graph / no-graph 0.999)
Relative WIS below 1.0 vs seasonal-naive, 2018-2019	0.545 overall, 0.66 at 4 weeks	met	
90% interval coverage of 80-95%	0.851 overall; 0.82-0.89 by horizon	met	
Access loss per municipality, two methods, four typhoons	network method, 1,641 municipalities, 4 storms	partly	raster (friction) method not wired in yet
Sensitivity bounds; Metro Manila benchmark	9 variants, $\tau \geq 0.99$; NCR 30% vs ~22%	met	
Dashboard for any historical or shifted track in under a minute	60 pre-computed scenarios; loads in about 6 s	met	
System Usability Scale from three target users	not done	not yet	needs the user sessions
Foundation-model benchmarks (Chronos-2, TabPFN-TS)	Chronos-2 script written, not run	not yet	needs the torch install

PROPOSAL CORRECTIONS FOUND IN THE DATA

- 126 dengue reporting units, not 81 provinces; deaths unreliable (2016 case fatality 3.9%); 2020 collapsed under COVID-19.
- An open subnational leptospirosis series exists (Project CCHAIN, 12 cities, 2008-2022).
- EMSR478 (Rolly) has damage grading only, no flood delineation; the Global Flood Database (Tellman et al., 2021) ends in 2018.
- Flood hazard maps come from the UP NOAH archive, not LiPAD; 11 units have no 25-yr map.
- HydroRIVERS (Lehner & Grill, 2013) places the Philippines in the Australasia file, not Asia.
- IBTrACS (Knapp et al., 2010) USA R34 is missing on up to 58% of fixes of the backtest storms; JMA R30 fills the gaps.
- roroph routes describe 2024-2026 operations with partly coarse port coordinates.

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LIMITATIONS AND NEXT STEPS

THREATS TO VALIDITY

- **Two evaluation years.** 2018-2019 contains one epidemic; skill and calibration may differ in quieter or COVID-era years.
- **Surveillance artefacts.** Reporting delays and under-reporting are in the target; forecasts predict reported, not true, cases.
- **Observed flood.** Sentinel-1 sees each place every 6-12 days and misses flooding under dense vegetation and in built-up areas; shifted tracks fall back to static hazard maps.
- **Network gaps.** OSM completeness is uneven outside Luzon; ferries are 2024-2026 routes; hospitals are an OSM / NOAH fallback without facility level or capacity.
- **Provisional rules.** Action-tier thresholds, road speeds and the cut-off rule are assumptions, partly addressed by the sensitivity analysis.
- **Causality.** PCMCI+ links are conditional dependencies, not transmission routes.

NEXT STEPS

1. Add a second radar or optical source between Sentinel-1 passes and compare observed-flood A with the hazard proxy.
2. Wire in the MAP friction raster method (Weiss et al., 2020) and report the network-vs-raster gap per province.
3. Recalibrate R and review the tiers with DOH and LGU users; run the System Usability Scale test.
4. Replace the hospital fallback with NHFR; validate pcodes against PSGC 2Q 2026.
5. Run the Chronos-2 benchmark (Ansari et al., 2025); extend to leptospirosis with the CCHAIN city series.

REPRODUCIBILITY

`setup.bat` (or `download_data.bat`) → `build.bat` → `run_app.bat`. This report and the slides are regenerated with `python scripts/make_results.py`. Missing inputs are skipped, listed in `data/processed/status.json` and on the dashboard; re-running after they arrive needs no code change.

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